



Numerical study of laminar premixed ammonia/air flames enriched with nitrogen oxides

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ABSTRACT

The interactions between NH_3 and nitrogen oxides (including N_2O and NO_x) offer unique insights into ammonia's combustion chemistry via active nitrogen–nitrogen coupling mechanisms. An in-depth understanding of these interactions is essential for the development of kinetic models, emission control, and strategies for enhancing combustion via modified oxidizers. This study presents a comprehensive investigation of laminar premixed ammonia/air flames enriched with N_2O or NO_x via one-dimensional simulations. The results demonstrate that all three nitrogen oxide species significantly enhance flame propagation, with NO showing the strongest enhancement at low concentrations, followed by NO_2 and N_2O . However, with increasing NO addition, the flame speed attains a maximum at $x_a \approx 0.6$ and then gradually declines with further enrichment. Such non-monotonic trends are not seen with N_2O or NO_2 addition. A decoupling analysis is further employed to isolate oxy-enrichment, thermal, chemical, and transport effects. For all three cases, the thermal effect is the dominant contributor. Interestingly, NO exhibits dual chemical behavior: it promotes combustion at low concentrations but inhibits it at high levels. The key reactions are also identified with the rate of progress and sensitivity analyses. Meanwhile, the correlations between the simulated laminar flame speeds and the theoretical expressions are examined over different addition ratios. Besides, pollutant emissions are also examined, showing increased NO levels with N_2O or NO_x addition. The chemical effects have a more pronounced impact on NO emissions than on flame speed. At the end, the flammability limits are determined where the radiative heat losses are modeled via the optically thin model. The results indicated a significant widening of the flammable range, with N_2O or NO_x addition. These findings provide critical insights into nitrogen–nitrogen coupling mechanisms and offer pathways for enhanced ammonia combustion technologies in carbon-free energy systems.

Novelty and significance

Understanding the dual role of nitrogen oxides as both pollutants and active participants in ammonia combustion remains a critical gap in low-carbon energy research. This study addresses this gap by offering a systematic quantification of oxy-enrichment, thermal, chemical, and transport effects induced by N_2O , and NO_x addition, in modifying flame speed and pollutant emissions of ammonia/air flames. It reveals a non-monotonic response of flame speed to NO addition ratio, which is not seen for N_2O or NO_2 additions. Another novelty lies in the quantification of correlations between simulated flame speeds and theoretical expressions, providing insight into the validity and limitations of simplified predictive models in ammonia systems. Moreover, this work presents the first estimations of flammability limit for $\text{NH}_3/\text{N}_2\text{O}/\text{NO}_x$ mixtures at both 1 atm and 5 atm. Together, these contributions advance fundamental understanding of $\text{NH}_3\text{-N}_2\text{O}$ and $\text{NH}_3\text{-NO}_x$ interactions, supporting the development of ammonia-fueled systems.

1. Introduction

Ammonia (NH_3) is a promising carbon-free fuel for decarbonizing power generation and transportation sectors [1,2]. Extensive research

has been conducted over the last two decades to explore its stable and clean combustion under various conditions [1,3,4]. Nevertheless, the majority of studies use O_2 as the sole oxidizer, with comparatively

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limited attention on nitrogen-containing oxidizers, e.g., nitrous oxide (N_2O) and nitrogen oxides (NO_x) [5,6].

The oxidation of NH_3 by N_2O and/or NO_x offers a valuable opportunity to study reactive nitrogen–nitrogen interactions and advance the understanding of NH_3 kinetics. These interactions are also highly relevant to practical combustion scenarios, e.g., exhaust gas recirculation, rich-lean staged combustion, and selective non-catalytic reduction (SNCR) [6,7]. In such environments, the nitrogen oxides can mix with ammonia. The localized N_2O or NO_x enrichment may lead to enhanced reactivity, potentially causing flashback or undesired coupling with upstream combustion zones. A detailed understanding of flame propagation under these conditions is thus essential for achieving effective NO_x reduction without compromising flame stability. Furthermore, nitrogen oxide–ammonia interactions are also critical in solid propulsion applications. Many solid rocket propellants and energetic materials contain carbon, hydrogen, oxygen, and nitrogen compounds [5]. Upon decomposition, these materials release nitrogen oxides and ammonia that can sustain a flame above the propellant surface and, in turn, influence the burning of the solid propellant. Consequently, understanding nitrogen oxide flame behavior is of fundamental importance across various combustion contexts.

Nitrous oxide is a potent greenhouse gas with a global warming potential 273 times that of CO_2 for a 100-year time scale [8]. In ammonia combustion, N_2O can form as an intermediate or byproduct, particularly under fuel-lean or low-temperature conditions [6]. Despite being viewed as a pollutant, N_2O is also a potential oxidizer, particularly in satellite propulsion applications [9], by acting as a precursor to atomic oxygen. Compared with air, N_2O can readily decompose into nitrogen and oxygen at elevated temperatures and pressures, releasing substantial energy [10]. Armjtag and Gray [10] conducted one of the earliest experiments on the flammability limits and flame speeds of various ammonia-containing mixtures nearly sixty years ago. It is documented that the N_2O addition broadens the flammability limits and enhances the laminar flame speed [11], which has also been confirmed by recent experiments [12,13]. The laminar flame speed of stoichiometric $\text{NH}_3/\text{N}_2\text{O}$ mixtures can reach up to 78 cm/s at 1 atm and 298 K, nearly ten times faster than stoichiometric NH_3/air flames. Furthermore, combustion supported by N_2O exhibits a greater propensity for flame acceleration and can potentially lead to deflagration-to-detonation transition [14,15].

In parallel, $\text{NH}_3\text{--NO}_x$ interactions have long been a focal point in the context of NO_x mitigation, particularly via the SNCR [7,16]. The ideal temperature range for SNCR is between 1120 and 1450 K. Accordingly, most experimental and modeling works to date have focused on this temperature regimes, with relatively limited exploration of high-temperature conditions [6,7]. However, practical combustion environments often operate at much higher temperatures, particularly in combustion enhancement scenarios, e.g., hydrogen blending and oxygen enrichment, where flame temperatures can exceed 2000 K [3]. Understanding $\text{NH}_3\text{--NO}_x$ interactions under such flame conditions is therefore crucial.

Early experiments on NH_3/NO mixtures [10,17,18] revealed that NH_3 begins to react with NO at lower temperatures than H_2 , and the resulting laminar burning velocity is roughly twice that of H_2/NO flames. In the case of NO_2 , the reaction with NH_3 occurs almost instantaneously upon mixing, making controlled flame speed measurements challenging. Accordingly, despite the available flame speed measurements for H_2/NO_2 or hydrocarbon/ NO_2 mixtures [17,19], corresponding data for NH_3/NO_2 flames remain scarce. Recently, the spherical flame has been employed to measure the laminar burning velocities of $\text{NH}_3/\text{NO}/\text{N}_2$ mixtures [20–23], providing new insights into these reactive systems. They show that using NO as the oxidizer enhances the flame propagation, and the enhancement is primarily driven by thermal effects [22]. In contrast, chemical effects flames tend to inhibit flame propagation, and the interplay between these two opposing effects can lead to non-monotonic variations of flame speed in $\text{NH}_3/\text{H}_2/\text{NO}$ mixtures [22].

Despite growing experimental and modeling efforts, systematic studies considering different nitrogen oxide species remain limited. The present work numerically examines the effects of N_2O , NO , and NO_2 on ammonia/air flames using the canonical steady laminar planar premixed flame. The laminar flame speed, a fundamental metric for characterizing flame propagation, is utilized to evaluate their impact on global flame dynamics. Furthermore, a decoupled analysis is performed to isolate the thermal, chemical, and transport contributions of these nitrogen oxides to both flame propagation and downstream pollutant emission. At the end, the impacts of N_2O and NO_x on the flammability limits are assessed. These findings, while derived from a simplified and idealized configuration, could support the understanding and predictive modeling of $\text{NH}_3/\text{N}_2\text{O}/\text{NO}_x$ mixtures in more complex combustion environments.

2. Numerical model

2.1. Steady planar propagating flame

One-dimensional planar premixed flames are simulated using the open-source software Cantera [24]. The diffusion flux is calculated using a multi-component model incorporating the Soret effect. Radiative heat transfer is only considered for flammability limit calculations in Section 3.3 with key information provided in Section 2.3. Without further specification, the ambient temperature and pressure are $T_u = 298$ K and $P_u = 1$ atm, respectively. In accordance with Refs. [25,26], the grid parameters are: normalized curvature of 0.1, normalized slope of 0.05, and spacing ratio of 2.0, leading to over 100 grid points. Grid independence has been achieved, as demonstrated in the supplementary material. The initial domain length is 20 cm, which can be adjusted to a greater length if it fails to sufficiently accommodate the flame in the original domain. Regarding the kinetic model, we collected over 700 experimental laminar flame speed measurements for $\text{NH}_3/\text{O}_2/\text{N}_2\text{O}/\text{NO}/\text{N}_2$ mixtures at atmospheric conditions from literature and evaluated the predictive performance of 18 recent kinetic models against this dataset. The Stagni2020 model [27], comprising 29 species and 203 reactions, is found to outperform the others and therefore adopted here. Comprehensive information on the experimental dataset, along with detailed model comparisons, is provided in the supplementary material and thereby not reiterated here. Moreover, additional simulations with different kinetic models confirm that, despite some quantitative variations, the overall trends in flame speed and pollutant emissions remain consistent, as provided in the supplementary materials.

2.2. Emission estimations

The emissions are evaluated from 5 mm downstream of the flame front (location of maximum heat release rate), which is consistent with common experimental practice, where exhaust gas is usually sampled at fixed probe positions. These emissions are then normalized to 15% O_2 concentration, following the standard practices for industry gas turbines [28]. Specifically, the adjusted concentration is computed using:

$$X_i @ \text{dry}, 15\%\text{O}_2 = X_i @ \text{dry} \times \frac{X_{\text{O}_2, \text{corr}} - 0.15}{X_{\text{O}_2, \text{corr}} - X_{\text{O}_2} @ \text{dry}} \quad (1)$$

where $X_i @ \text{dry}, 15\%\text{O}_2$ denotes the adjusted concentration normalized to 15% oxygen on a dry basis, $X_i @ \text{dry}$ is the measured dry-basis concentration, and $X_{\text{O}_2} @ \text{dry}$ the measured dry-basis oxygen concentration in the exhaust. Since the nitrogen oxide species introduce additional oxygen into the system, a corrected oxygen concentration is considered, i.e., $X_{\text{O}_2, \text{corr}}$. It accounts for the total oxygen content in the oxidizer stream by assuming that all nitrogen-containing oxidizer fully decomposes into O_2 and N_2 .

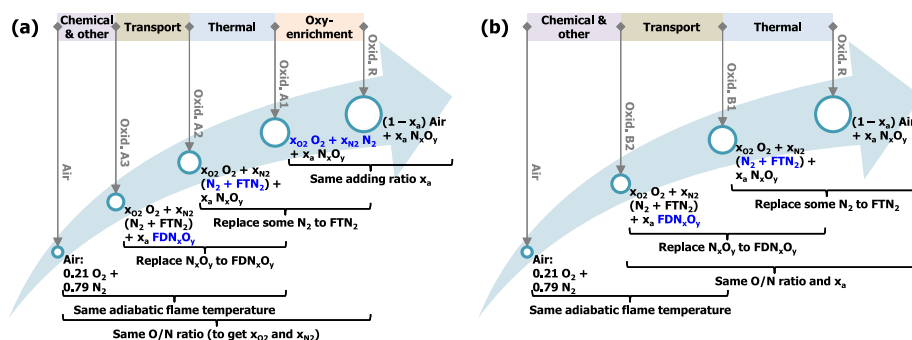


Fig. 1. Illustration of the decoupling method to isolate the individual effects associated with reactive oxidizer additions: (a) with and (b) without oxy-enrichment effects separated.

Additionally, another commonly adopted approach is based on the post-flame residence time. This is particularly useful for reduced-order modeling, since it reflects the effective reaction duration within practical combustors. Our comparisons between the distance-based and residence-time-based approaches show that, while the quantitative values (notably NO under lean conditions) can vary slightly, the overall qualitative emission trends remain consistent.

2.3. Determination of flammability limit

The flammability limit of a gaseous mixture, defined under specific temperature and pressure conditions, refers to the propagation boundary of a planar unstretched premixed flame [29,30]. This limit is determined by the chemical kinetics and thermal radiation. In this study, we adopt the arc-length continuation method [31,32] implemented in Cantera [24] to calculate flame solutions across a full range of equivalence ratios and identify the corresponding extinction boundaries [32].

Recent experiments by Xia et al. [33] have demonstrated that the optically thin model provides a reasonable approximation of total flame radiation intensity under atmospheric and oxygen-enriched ammonia combustion scenarios. Therefore, we calculate the radiative heat losses using the same model. Four radiative species, including H₂O, NH₃, NO, and N₂O, are considered, with the Planck-mean absorption coefficients from Ref. [34]. Note that radiation is considered only for discussions on flammability limits.

2.4. Initial oxidizer condition and decoupling method

In the present study, N₂O, NO, or NO₂ are introduced into the air stream. These nitrogen-containing oxidizers can be collectively represented by the general formula N_xO_y, and the corresponding global reaction with NH₃ is written as: $4y \text{ NH}_3 + 6 \text{ N}_x\text{O}_y \rightarrow (3x + 2y) \text{ N}_2 + 6y \text{ H}_2\text{O}$. The enrichment ratio x_a is defined as the molar fraction of N_xO_y in the oxidizer stream, i.e., $(1 - x_a) \text{ air} + x_a \text{ N}_x\text{O}_y$. The initial unburnt premixtures are prepared by blending ammonia with the oxidizer stream, with the overall composition determined by the equivalence ratio ϕ , which is evaluated based on the elemental molar fractions of hydrogen and oxygen. In this study, the final combustion products are assumed to consist of N₂ and H₂O, with possible residual NH₃ or O₂ depending on the equivalence ratio. The simulations are conducted over a range of equivalence ratios ($\phi = 0.6$ to 1.8), and enrichment ratios ($x_a = 0$ to 1.0). Note that the x_a values employed here may exceed typical post-flame NO_x concentrations in practical combustors. These higher concentrations are used primarily to amplify the effects of N_xO_y on NH₃ flame, thereby providing clearer insights into underlying mechanisms. Nevertheless, the study also includes lower concentrations (e.g., $x_a = 0.01$), which are more relevant to practical post-flame conditions. At these dilute concentrations, however, the effects of N_xO_y are diminished.

To isolate the various effects related to N_xO_y addition, several artificial species are introduced in simulations, following Refs. [35,36]. Specifically, the species FTN₂ mimics the transport properties of N₂, but its specific heat capacity is set to be 100 times higher than that of N₂. As a result, even small additions of FTN₂ can significantly lower the adiabatic flame temperature, thereby isolating the thermal effects associated with the elevated flame temperature. Besides, given its trace-level presence (e.g., < 2%), FTN₂ is assumed to be chemically inert and exerts negligible influence on the overall reaction kinetics. Another artificial species, FDN_xO_y, is constructed to replicate the thermodynamic and chemical properties of its corresponding N_xO_y species, while adopting the same diffusion coefficients as N₂. This enables a decoupled assessment of transport effects.

Fig. 1(a) presents a schematic of the decoupling strategy to isolate the individual effects associated with N_xO_y addition. Four major effects are considered: oxy-enrichment, thermal, transport, and chemical interactions & others. The oxy-enrichment effect refers to the increase of oxygen elemental molar fraction relative to air in the oxidizer stream due to the introduction of N_xO_y. To isolate this effect, the O₂ and N₂ concentrations in the oxidizer stream are adjusted to match the O/N ratio of air, thereby keeping the overall oxygen elemental content constant. This yields an adjusted oxidizer stream referred to as **Oxid. A1**. Moreover, the thermal effect stems from the enthalpy increase or heat capacity changes due to N_xO_y addition, which typically leads to an increase in adiabatic flame temperature. To counteract this and isolate the thermal contribution, a small amount of FTN₂ is introduced to replace part of the N₂ via an iterative approach, thereby reducing the adiabatic flame temperature to that of the air, resulting in **Oxid. A2**. The thermal impact can then be analyzed by comparing the results between **Oxid. A1** and **A2**. The transport effect arises due to differences in the molecular transport properties between the nitrogen oxide and the background gases. To evaluate this, the N_xO_y species are replaced with their analogs FDN_xO_y. The associated oxidizer stream is referred to as **Oxid. A3**. Ultimately, any residual differences from the reference air stream to **Oxid. A3** are attributed to chemical interactions and other secondary effects.

In the above comparisons, the oxy-enrichment effect is isolated by varying the elemental nitrogen-to-oxygen ratio in the oxidizer stream. Consequently, there exists an upper limit of x_a in assessing such effects, which is uniquely determined by the elemental molar fraction of oxygen in N_xO_y. Furthermore, the modification from **Oxid. R** to **Oxid. A1** inevitably introduces additional thermal and chemical effects compared to the reference cases. To further account for these thermal and chemical effects related to the oxy-enrichment, an alternative decoupling method is employed to separate the net thermal, transport, and chemical effects, as illustrated in Fig. 1(b). Specifically, the **Oxid. B1** and **Oxid. B2** are introduced by selectively modifying portions of nitrogen or N_xO_y. The underlying idea remains consistent with previous approaches and is therefore not reiterated in detail.

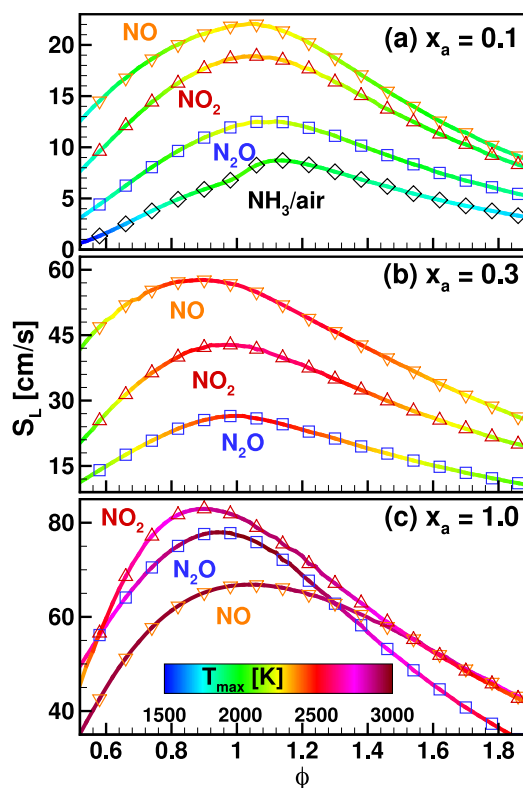


Fig. 2. Laminar flame speeds of $\text{NH}_3/\text{N}_x\text{O}_y/\text{air}$ mixtures at: (a) $x_a = 0.1$, (b) $x_a = 0.3$, and (c) $x_a = 1.0$. Lines are colored by the respective maximal flame temperature T_{max} . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Results and discussion

3.1. Laminar flame speed

3.1.1. Influence of oxidizer and x_a

Fig. 2 presents the laminar flame speeds of $\text{N}_2\text{O}/\text{NO}_x$ -enriched ammonia flames at various equivalence ratios. Apparently, the addition of nitrogen oxide, i.e., N_2O , NO , and NO_2 , to the air stream markedly enhances the flame speed and the flame temperature (visualized by the color of the curves). It is shown from Fig. 2(a) that N_2O addition with $x_a = 0.1$ leads to approximately 50% increase in flame speed compared to the NH_3/air flame. Such enhancement becomes more pronounced with the addition of NO or NO_2 at the same x_a , approximately yielding an increase of 1.5 or 1.2 times, respectively. Additionally, the N_xO_y enrichment shifts the peak laminar burning velocities toward the leaner side: from around $\phi = 1.12$ for NH_3/air to approximately 1.05 for N_2O -, NO -, and NO_2 -enriched flames at $x_a = 0.1$. Similar phenomena have been reported in Ref. [21]. An asymmetric change in flame speed with equivalence ratios is also noted, with the burning velocity rising more swiftly on the lean side, while changes on the rich side are more gradual. This trend is particularly obvious for NH_3/NO_2 mixtures, which may be related to the decomposition of these nitrogen oxides [17].

As x_a increases to 0.3 in Fig. 2(b), the enhancement effect becomes stronger, both in terms of flame speed and temperature. The equivalence ratios corresponding to the peak laminar burning velocities also lie at the leaner mixtures, approximately $\phi = 1.00$ for N_2O , 0.90 for NO , and 0.92 for NO_2 . It is also observed that NO remains the most effective additive among the three oxidizers, while N_2O is the least effective. However, as x_a increases to 1.0 (i.e., without air in the oxidizer stream), a reversal of the trend is observed in Fig. 2(c). Specifically, at lean and near-stoichiometric conditions, the flame speed of NH_3/NO mixtures

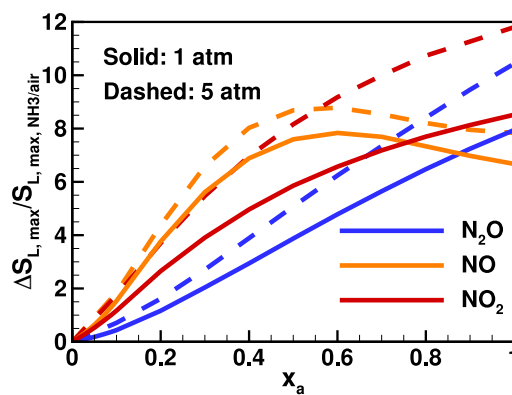


Fig. 3. Change in the maximum laminar flame speeds relative to the baseline NH_3/air flame for N_2O -, NO -, and NO_2 -enriched NH_3/air flames at different x_a .

becomes lower than that of $\text{NH}_3/\text{N}_2\text{O}$ and the NH_3/NO_2 mixtures. On the rich side, due to the abundance of fuel, this effect is mitigated. At $\phi = 1.5 - 1.8$, for example, NO exhibits superior performance compared to N_2O , while remaining only comparable with NO_2 . This phenomena has not been observed at lower x_a , e.g., in Figs. 2(a)–(b).

The change in the maximum laminar flame speeds relative to the baseline NH_3/air flame serves as a useful indicator for quantifying the enhancement effects of N_xO_y additions. Fig. 3 illustrates the relative change in the maximum laminar burning velocities ($\Delta S_{L, \text{max}}/S_{L, \text{max, NH}_3/\text{air}}$) for N_2O -, NO -, and NO_2 -enriched NH_3/air flames across various enrichment ratios. At small enrichment levels, e.g., $x_a < 0.4$, a roughly linear increase in the enhancement factor is observed. Among three oxidizer additives, NO exhibits the strongest enhancement effect, with the fitted slope being approximately 1.4 times that of NO_2 and twice that of N_2O . As x_a increases, N_2O continues to show an approximately linear trend up to $x_a = 1.0$, while the effect of NO_2 gradually diminishes for $x_a > 0.6$. Interestingly, NO displays a non-monotonic trend: there appears to be an optimal enrichment ratio around $x_a = 0.6$, at which flame speed enhancement is maximized. Beyond this point, further increases or decreases in NO concentration result in reduced flame speeds. This is likely due to the chemical interactions between NH_3 and NO . Meanwhile, Fig. 3 also presents results at 5 atm. Elevated pressure amplifies the effect of N_xO_y enrichment on flame speed, resulting in greater relative changes compared to atmospheric conditions. Nonetheless, the qualitative trends remain consistent, supporting the robustness of our conclusions under higher pressure.

3.1.2. Physical–chemical effects on S_L

To quantify the physical–chemical effects associated with reactive oxidizer addition, a fictitious gas approach is employed to decouple the various contributing factors, as detailed in Section 2.4. The corresponding results are presented on Fig. 4. It has been noted that the oxy-enrichment considerably influences the laminar burning velocity across all three nitrogen oxides, whereas the effects of transport are minimal, aligning with findings in Refs. [35,36].

Specifically, for the N_2O addition, the thermal effect is the most dominant contributor to the flame speed enhancement, while the oxygen enrichment plays a secondary role. Interestingly, the chemical effect appears to suppress flame propagation slightly. For NO addition, the oxy-enrichment becomes more pronounced. Notably, the thermal contribution does not increase at high NO concentrations, e.g., $x_a > 0.3$. This is attributed to incomplete NO reactions. It has been noted that the flame temperature under this condition is also lower than the adiabatic flame temperature estimated by assuming complete conversion to N_2 and H_2O . The chemical and other effects then constitute the third major influence on flame propagation: they tend to enhance the laminar

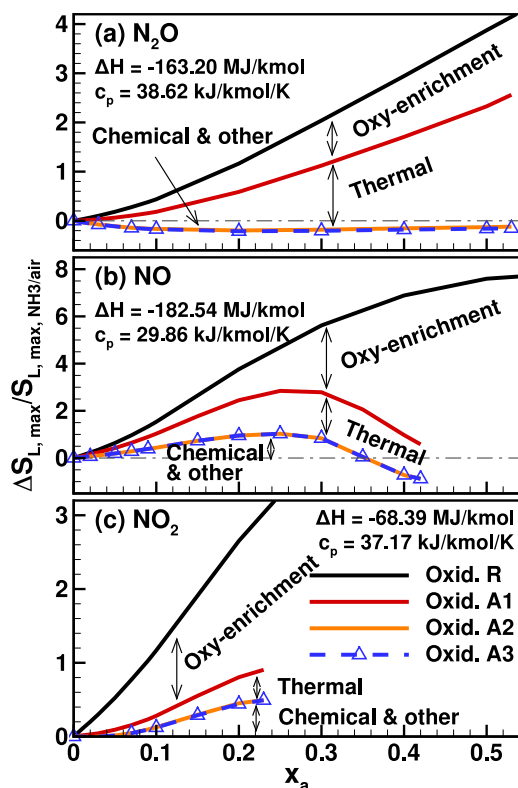


Fig. 4. Change of $\Delta S_{L,max}/S_{L,max,NH_3/air}$ for N_2O -, NO -, and NO_2 -enriched NH_3/air flames by employing the decoupling approach detailed in Fig. 1(a). ΔH represents the enthalpy changes associated with the N_xO_y decomposition reaction i.e., $2 N_xO_y \rightarrow x N_2 + y O_2$, and c_p denotes the partial molar specific heat capacities at constant pressure. Both are evaluated at 1 atm and 298 K.

burning velocity at lower NO concentrations but may inhibit flame propagation at higher enrichment levels. With NO_2 , oxygen enrichment is the most influential factor, while thermal and chemical effects contribute comparably but to a lesser extent. Importantly, among the three oxidizers, only NO_2 exhibits a consistently positive chemical effect across all enrichment levels, highlighting its unique role in promoting flame propagation.

These observations suggest that while the relative importance of oxygen enrichment, thermal, and chemical effects varies among the oxidizers, thermal effects always play a central role in influencing the flame speed. The thermal effects are primarily governed by two critical parameters: the enthalpy increase resulting from the addition of nitrogen oxide and the changes in heat capacity (c_p). These two parameters are evaluated at 1 atm and 298 K, and presented in Fig. 4. It is shown that NO yields the highest enthalpy increase and the lowest heat capacity, which facilitates a greater temperature increase. Consequently, the thermal effect of NO addition is the strongest among the three species. However, at higher NO concentrations, the enhancement effect becomes limited due to the incomplete reaction of NO . In contrast, NO_2 exhibits the weakest thermal enhancement, which is reasonable given its relatively high heat capacity (comparable to that of N_2O) and its lower enthalpy increases.

The thermal and chemical effects associated with oxygen enrichment are also considered using the method illustrated in Fig. 1(b), and the corresponding results are presented in Fig. 5. It is observed that under all conditions, the thermal effect, primarily resulting from the increased enthalpy of the oxidizer stream, is dominant in the enhancement of flame speed. In contrast to thermal contributions, the transport effects are minimal, likely because the transport properties of these nitrogen oxide species are already similar to those of the

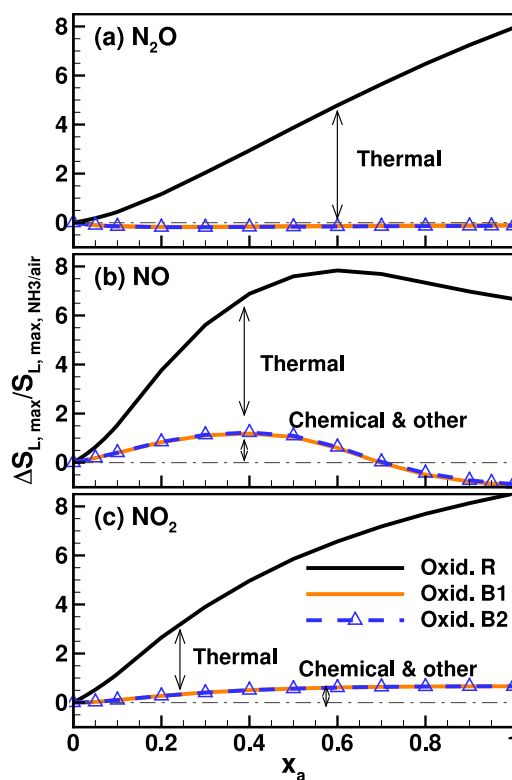


Fig. 5. Change of $\Delta S_{L,max}/S_{L,max,NH_3/air}$ for N_2O -, NO -, and NO_2 -enriched NH_3/air flames by employing the decoupling approach detailed in Fig. 1(b).

background gas. Although only the $\Delta S_{L,max}$ is presented here, similar conclusions hold at fixed equivalence ratios (i.e., ΔS_L at $\phi = 0.6$, 1.0, and 1.4), and are therefore not shown for brevity. Regarding the chemical effects, both NO and NO_2 demonstrate a more significant impact on flame propagation, while N_2O has a minimal chemical effect, albeit a negative one. This observation aligns with the data presented in Fig. 4. Besides, the non-monotonic variation of flame speed with x_a in $NH_3/NO/air$ mixtures in Fig. 3 is primarily attributed to chemical effects. This is supported by three key observations: first, the influence of transport effects is negligible, as also shown in prior studies [26, 35]; second, the flame temperature increases monotonically with x_a , indicating consistently positive thermal contributions; and third, the observed non-monotonic behavior must originate from the remaining effects, i.e., the chemical effects. This is further corroborated by results shown in Figs. 4(b) and 5(b).

3.1.3. Chemical kinetics analyses

This subsection focuses on the chemical effects induced by the addition of N_xO_y . The key reactions directly responsible for the consumption and production of N_2O , NO , and NO_2 are examined first in the respective addition scenarios. Fig. 6 shows the normalized contributions of dominant reactions for each oxidizer at stoichiometric conditions, i.e., $\phi = 1.0$. These reactions collectively account for more than 80% of the total consumption. Note that similar trends are observed at lean ($\phi = 0.7$) and rich ($\phi = 1.4$) conditions and are thereby omitted here for brevity. Besides, Fig. 6 also include results at 5 atm. It is observed that elevated pressure has a negligible influence on the ranking and relative importance of the key reactions. Given the strong similarity between the results at 1 atm and 5 atm, only the atmospheric pressure data are presented in the subsequent analysis for clarity and conciseness.

Specifically, in N_2O -enriched flames, N_2O is primarily consumed through reactions R189 and R187. The former one, i.e., R189, involves

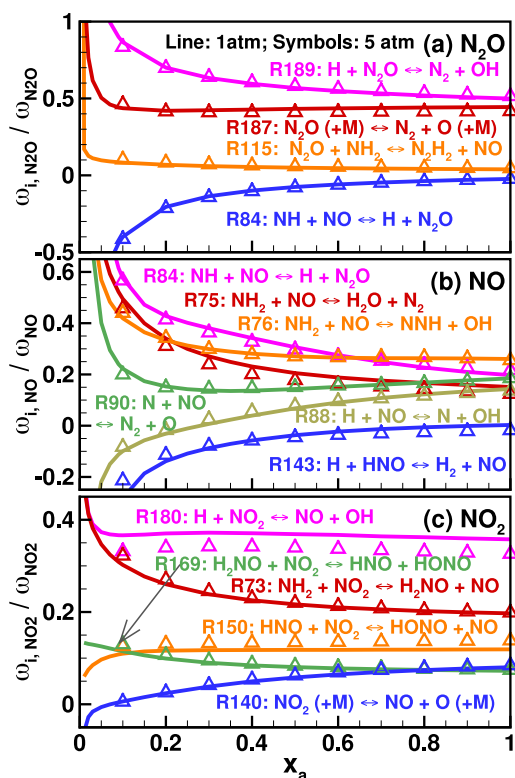


Fig. 6. Relative contribution of key reactions accounting for the direct consumption/production of these nitrogen oxide species for stoichiometric premixed (a) N_2O -, (b) NO -, and (c) NO_2 -enriched NH_3 /air flames. The $\omega_{i, \text{species}}$ denotes the consumption rate of a given species through reaction i , and ω_{species} the cumulative consumption rate of that species.

one H radical, which facilitates the rapid decomposition of N_2O and promotes the formation of OH, thereby enhancing flame reactivity. The latter reaction R187 is a third-body reaction which aids the decomposition of N_2O primarily through the involvement of H_2O , N_2 , and O_2 . Additionally, at low N_2O addition ratios, there is still a significant formation of N_2O via the $\text{NH} + \text{NO}$ pathway (R84). However, this pathway becomes less prominent at higher N_2O concentrations, where direct decomposition reactions dominate.

NO addition presents a more complex chemical scenario, with a wide array of reactions involved. Specifically, 41 reactions directly participate in NO consumption or formation, compared to only 14 for N_2O and 37 for NO_2 . This underscores the chemically active role of NO in flame propagation, where it not only participates in key reaction pathways but also significantly alters the radical pool, thereby affecting flame structure and speed. It is shown that at low x_a , R84 is the dominant NO consumption pathway, producing one H radical and N_2O . The resulting N_2O is subsequently converted into N_2 via R189 and R187, as discussed earlier. Another important set of interactions involves NO and NH_2 , where two reaction channels coexist, i.e., R75 and R76. The reaction R75 leads to the formation of H_2O and N_2 , while R76 produces NNH and OH. As x_a increases, the reaction R76 tends to be more reactive compared to R75. Additionally, reactions R88 and R90 also exhibit increased activity at higher x_a . One interesting observation is that R88 displays a concentration-dependent directionality: at low NO concentrations, it proceeds toward NO formation, whereas at high NO concentrations, it reverses and contributes to NO consumption. Regarding the NO formation, R143 involving HNO is particularly important. It not only contributes to NO production but also facilitates NH_3 pyrolysis, leading to H_2 formation.

For NO_2 , the primary consumption pathway is R180, which, like R189 for N_2O , entails one H radical and results in the formation of

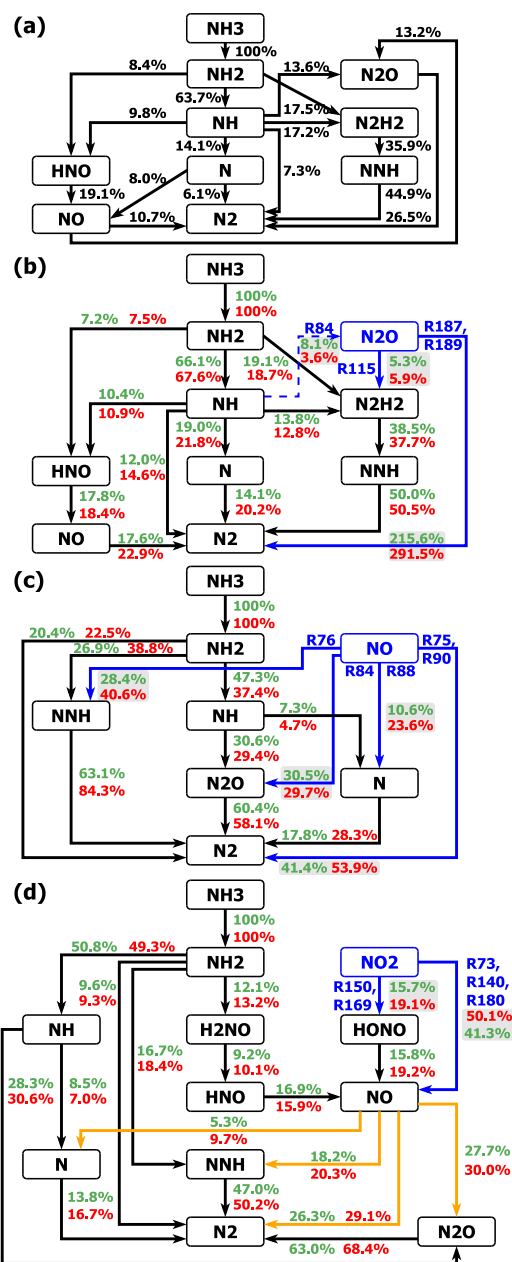


Fig. 7. Main reaction networks following N element in stoichiometric premixed (a) NH_3 /air, (b) $\text{NH}_3/\text{N}_2\text{O}$ /air, (c) NH_3/NO /air, and (d) NH_3/NO_2 /air flames. The normalized reaction fluxes for $x_a = 0.0$ (black), 0.5 (green), and 1.0 (red) are annotated next to the arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

one OH radical. However, unlike N_2O , NO_2 does not undergo significant unimolecular decomposition. It is shown that even at $x_a = 1.0$, the contribution from third-body reaction is still below 0.1. Instead, NO_2 consumption relies more on reactions with nitrogen-containing intermediates, such as NH_2 and HNO. This underscores the stronger coupling between NO_2 chemistry and NH_3 oxidation compared to the more weakly interacting N_2O . Besides, the typical product of the NO_2 reaction is NO except for the R169 in Fig. 6. Therefore, it may indirectly affect flame behavior through the pathway for NO reactions.

Fig. 7 further illustrates the main reaction networks following nitrogen element for stoichiometric premixed NH_3 /air flames without and with N_xO_y enrichment. The key reactions responsible for the

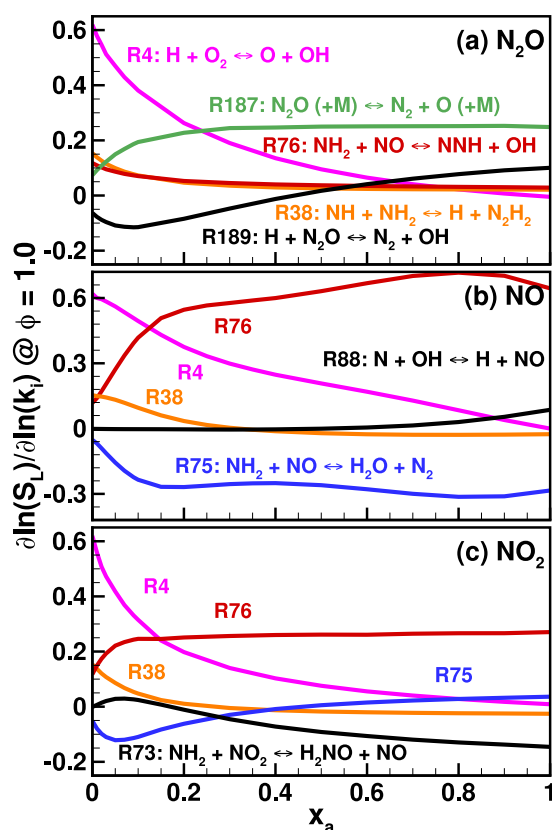


Fig. 8. Normalized sensitivities of S_L for the top 5 most important reactions of stoichiometric premixed (a) N_2O -, (b) NO -, and (c) NO_2 -enriched NH_3 /air flames.

consumption of N_xO_y are highlighted in blue. It is observed that for N_2O addition, the dominant NH_3 reaction pathways remain largely unchanged, with NH , HNO , and NNH being the key ammonia consumption routes. This is consistent with the results presented in Figs. 4(a) and 5(b). The N_2O influences the flame primarily through the consumption of H radicals, as also observed in Ref. [37]. This reduces the availability of H for direct reactions and diverts NH_2 and NH to react with other species such as OH , NH , and NO instead of H . Furthermore, the presence of N_2O enhances the $NH + NO \rightarrow N_2 + OH$ pathway, while suppressing the competing $NH + NO \rightarrow N_2O + H$ channel. As for NO and NO_2 addition, the reaction pathways of NH_3 undergo significant changes, primarily driven by interactions between NO_x and NH_i radicals. In the case of NO enrichment, both the HNO and N_2H_2 pathways are strongly suppressed. This suppression arises because a substantial fraction of NH_2 reacts with NO , which enhances the NNH pathway while reducing the contribution of the NH route. The remaining NH predominantly reacts with NO to form N_2O and H , leaving negligible HNO formation. Meanwhile, NH_2-NH and $NH_2-N_2H_2$ reactions are inhibited, further contributing to the disappearance of the N_2H_2 pathway, shown in Fig. 7(c). In contrast, NO_2 enrichment promotes the HNO pathway. But unlike in the NH_2 /air case, the HNO is formed primarily through the $NH_2 (+NO_2) \rightarrow H_2NO \rightarrow HNO$ route. Since NO_2 consumption predominantly yields NO , the NNH and N_2O pathways, driven by reactions between NH_i and NO , are also strengthened. Note that in the latter two scenarios, i.e., NO and NO_2 enriched flame, NNH is no longer formed via the N_2H_2 route as it is in the NH_3 /air situation. Instead, it is mainly formed through reactions between NH_i radicals and NO . In summary, the above results clearly demonstrate that NO and NO_2 additions exert more significant chemical effects on the flame, compared to N_2O addition.

To better identify the key reactions governing the laminar flame speed, a detailed sensitivity analysis of the laminar flame speed is performed. Fig. 8 presents the normalized sensitivities of the flame speed with respect to individual reaction rate constants, where only the top five most influential reactions are presented. It is shown that in all cases, an increase in the x_a leads to a marked decrease in the sensitivity of R4, which is attributed to the reduced O_2 content in the oxidizer stream.

For each reactive oxidizer, distinct reactions emerge as dominant contributors to flame propagation. In the case of N_2O -enriched flames, reaction R187 emerges as an important promoter of flame speed as the oxidizer addition ratio x_a increases from 0 to 0.2, beyond which its influence plateaus. A particularly notable observation is the behavior of reaction R189 in Fig. 8(a), which exhibits a dual effect: it inhibits flame propagation at low x_a but promotes it at higher x_a . This may be attributed to its competition with R4, a key chain-branching reaction. As R4's influence diminishes due to reduced O_2 , R189 becomes more favorable in providing the OH radicals for ammonia oxidation.

For NH_3/NO /air flames, the most important promoting reaction is R76, which converts NH_2 and NO into NNH and OH . The sensitivity of this reaction initially increases with x_a but subsequently decreases for $x_a \geq 0.8$, indicating a non-linear influence on flame propagation. In contrast, the most significant inhibitory reaction is R75, which converts NH_2 and NO into H_2O and N_2 . Notably, R75 is an exothermic reaction, and if its thermal contribution is excluded, the net chemical inhibition it imposes on flame speed would be even more pronounced. Besides, for $x_a > 0.3$, there is a sharp reduction in the contribution of R4, with no compensating reactions showing increased positive sensitivity to promote the flame. Consequently, R75 tends to play an increasingly important role under high- x_a conditions, and is responsible for the non-monotonic behavior of flame speed with NO addition. These results also highlighted critical roles of NH_2 - NO interactions in shaping flame behavior, particularly via the competition between R75 and R76. Additionally, reaction R88 shows a shift in behavior with increasing x_a : at low x_a , it proceeds in the direction of NO formation but has minimal impact on flame speed, whereas at high x_a , it proceeds in the reverse direction, consuming NO and producing OH , which slightly promotes flame propagation.

When it comes to NH_3/NO_2 /air flames in Fig. 8(c), the key reactions determining flame speed are largely similar to those observed in NH_3/NO /air flames, suggesting that NO_2 addition influences flame propagation in a more indirect manner. Compared to the NO -enriched cases, R75 becomes less significant. Notably, the sensitivity of R75 changes sign as x_a increases, and it tends to promote flame propagation at high x_a , which is different from the NH_3/NO /air flames. Additionally, it is also observed that the reaction R73, i.e., the second most important NO_2 consumption reaction, shows a negative influence on flame speed for $x_a \geq 0.3$, implying a chemically inhibitory role under these conditions. However, most of the direct reactions involving NO_2 do not appear in the top five most influential reactions, reinforcing the view that NO_2 acts more indirectly in modifying flame behavior. Based on the above discussions, it can be concluded that the enhancement of flame speed due to the addition of NO_2 is mainly via reaction R76.

3.1.4. Correlation of S_L

Another commonly adopted methodology to quantify flame speed enhancement, e.g., due to fuel blending or oxygen enrichment, is based on the theoretical expression for laminar flame speed [26,35]:

$$S_L \propto (\alpha Le)^{0.5} \exp\left(-\frac{E_a}{2RT_{ad}}\right), \quad (2)$$

where α is the thermal diffusivity, Le the Lewis number, E_a the activation energy, R the universal gas constant, and T_{ad} the adiabatic flame temperature. The first term on the right-hand side, $(\alpha Le)^{0.5}$, captures the contribution from preferential diffusion, while the exponential term reflects the combined kinetic and thermal effects. While the theoretical

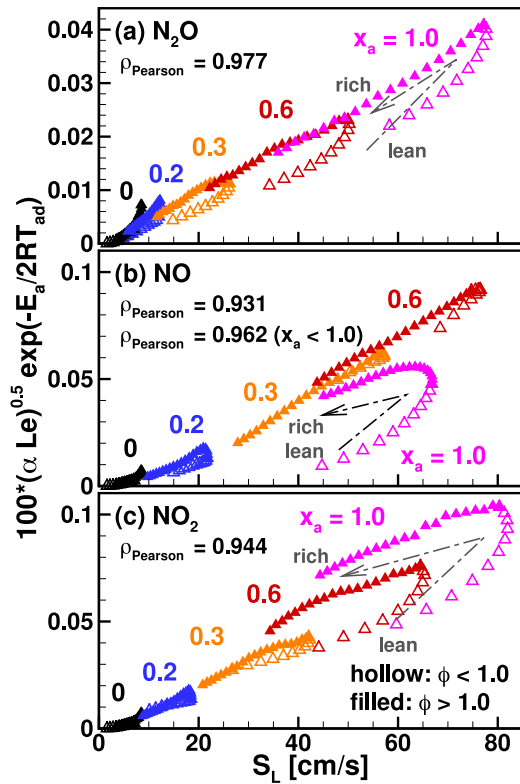


Fig. 9. Correlations between the simulated laminar flame speed and the theoretical expressions for (a) N_2O , (b) NO , and (c) NO_2 -enriched NH_3 /air flames.

correlation is well-established, its direct applicability to flame speed predictions in systems involving nitrogen oxide, such as N_2O , NO , and NO_2 , has not been systematically assessed. To address this, a comprehensive assessment is performed to evaluate the correlations between simulated laminar flame speeds and theoretical predictions. The evaluation of key parameters, including α , Le , E_a , and others, follows the methodology outlined in Ref. [35]. Specifically, E_a was determined by perturbing the inert composition N_2 within a range of -0.5% to $+0.5\%$. More details could be found in Ref. [35] and are therefore not repeated here. Note that the volumetric weighting method is utilized to assess the Lewis number of the oxidizer, taking into account both O_2 and N_xO_y in the oxidizer stream.

Fig. 9 shows the correlations between the simulated laminar flame speeds and the theoretical predictions across different oxidizer addition ratios. The Pearson correlation coefficient, $\rho_{Pearson}$, is also calculated using the data for $x_a = 0-1.0$ and $\phi = 0.6-1.8$, as displayed at the top left of each subfigure. It is shown that for N_2O , a strong and nearly linear positive correlation is observed across the entire range of x_a , indicating that the theoretical model effectively captures the combined thermal, transport, and chemical influences for this oxidizer. In the case of NO , a good agreement is found for $x_a \geq 0.6$. However, at $x_a = 1.0$, notable deviations emerge, particularly under fuel-lean conditions where NO concentrations are high. When the data at $x_a = 1.0$ is excluded, the $\rho_{Pearson}$ improves from 0.931 to 0.962, suggesting that the poorer fit at high NO enrichment significantly affects the overall trend. These discrepancies are likely due to the increasing influence of incomplete NO reactions that are not fully accounted for by the simplified theoretical framework. For NO_2 , the overall correlation remains reasonably strong, with a Pearson coefficient of $\rho_{Pearson} = 0.944$. However, noticeable discrepancies emerge at both the lean and rich extremes of the equivalence ratio range, particularly for $x_a \geq 0.6$. These deviations may stem from asymmetrical thermal or chemical behavior

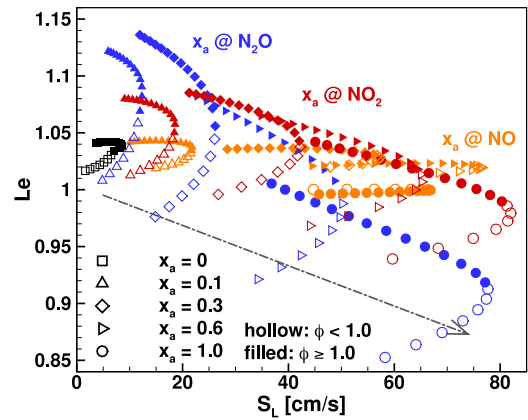


Fig. 10. Change of the effective Lewis number.

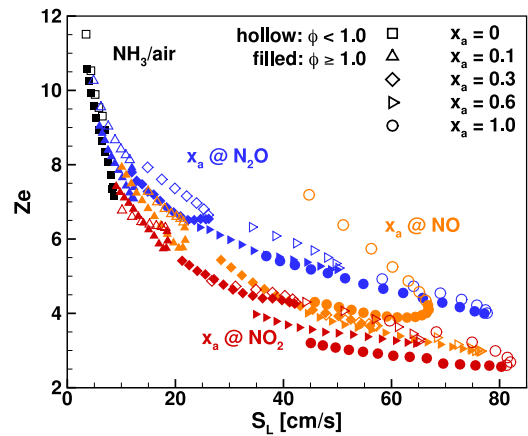


Fig. 11. Change of the Zeldovich number, i.e., $Ze = E_a(T_{ad} - T_u)/RT_{ad}^2$.

introduced by the additional oxidizer species at lean and rich sides. It is noted that in Eq. (2), the latter exponential term plays a dominant role, and this term has an exponential dependence on the E_a . In systems with complex reaction networks, characterizing kinetic behaviors with a single lumped parameter E_a is challenging. Consequently, this would lead to observable discrepancies between theoretical predictions and detailed simulation results.

Fig. 10 illustrates the change of the effective Lewis number (Le) across different equivalence ratios and oxidizer addition ratios. Generally, Le tends to be slightly above unity under fuel-rich conditions and slightly below unity under lean conditions. Despite these trends, the effective Lewis number remains close to 1.0 in all cases, indicating that transport effects on laminar flame speed are relatively minor. In particular, NO -enriched flames exhibit minimal variation in Le —especially in NH_3/NO mixtures, where Le is almost identical to unity across the entire equivalence ratio range. This is consistent with the similar transport properties of NO and N_2 . In contrast, N_2O addition leads to a more significant change of Lewis number, particularly at high N_2O levels. For instance, at $x_a = 1.0$, the Le falls below 1.0 for $\phi \leq 1.7$, and further drops to 0.85 at $\phi = 0.6$. This indicates an increased susceptibility to thermo-diffusive instabilities for N_2O enriched ammonia/air flames, particularly under lean conditions. Regarding NO_2 , it demonstrates an intermediate behavior between NO and N_2O , with moderate changes in Le across different equivalence ratios. Besides, similar to N_2O , higher NO_2 concentrations can also reduce Le , though the effect is less pronounced.

The Zeldovich number (Ze) provides a quantitative measure of the activation energy of a chemical reaction and serves as a key indicator

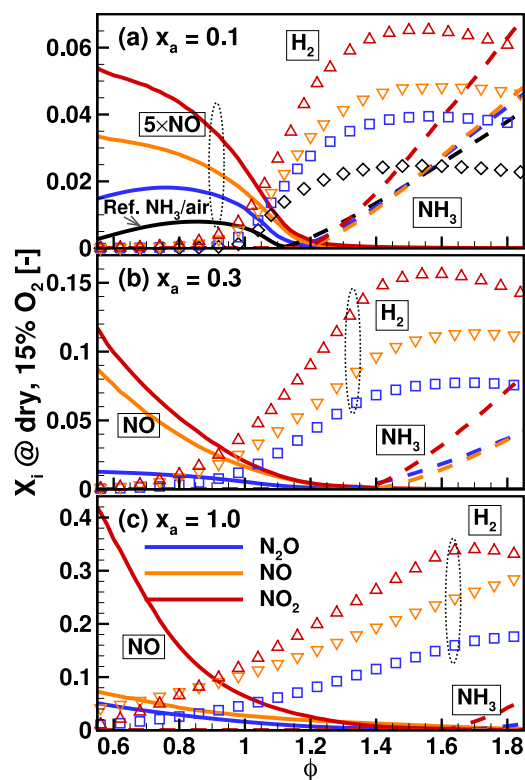


Fig. 12. Change of corrected NO (solid lines), NH_3 (dashed lines), and H_2 (symbols) concentrations in the post-flame zone with respect to ϕ for $\text{NH}_3/\text{N}_x\text{O}_y/\text{air}$ mixtures: (a) $x_a = 0.1$, (b) $x_a = 0.3$, and (c) $x_a = 1.0$.

of a flame's sensitivity to temperature perturbations. Flames with high Zeldovich numbers are generally more (less) prone to thermo-diffusive instabilities, when coupled with mixtures with sub-unity (super-unity) effective Lewis numbers [38]. Fig. 11 shows the variation of Ze under different mixture conditions. The Ze generally decreases near stoichiometric equivalence ratios or with the increase of x_a , except for NO at high concentrations. In particular, the Ze of lean NH_3/NO flames with $x_a = 1.0$ is significantly larger compared to the rich cases or $\text{NH}_3/\text{NO}/\text{air}$ flames with $x_a = 0.6$. Additionally, when comparing different oxidizers, it is observed that NO_2 is the most effective in reducing Ze , while N_2O performs the worst. This trend correlates with the positive chemical effects of NO_2 addition and the negative chemical effects of N_2O addition, as shown in Section 3.1.2.

3.2. Pollutant emissions

In this section, we investigate the effects of N_xO_y addition on the formation of nitrogenous pollutants and unburned NH_3 and H_2 . The analysis is based on post-flame species profiles evaluated 5 mm downstream of the flame front. Over an equivalence ratio range of 0.55–1.85, the concentrations of N_2O and NO_2 in the post-flame region are significantly lower compared to NO and NH_3 and are therefore omitted from the figures presented below for clarity. It should be noted, however, that N_2O emissions increase notably near the flammability limits, especially under lean conditions, as illustrated in the supplementary material.

Fig. 12 presents the extracted emissions of NO, NH_3 , and H_2 , as functions of ϕ . Consistent with the behavior observed in NH_3/air flames (black lines and symbols), NO emissions are high under lean conditions, while NH_3 and H_2 concentrations increase at fuel-rich conditions. With increasing x_a , both NO and H_2 emissions rise, whereas residual NH_3 declines due to enhanced thermal decomposition of NH_3 into H_2 .

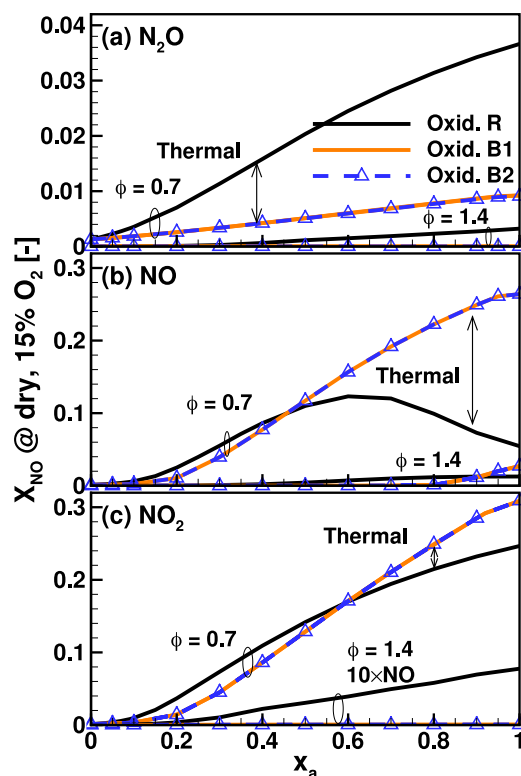


Fig. 13. Changes of corrected NO concentrations in the post-flame zone with respect to x_a for (a) N_2O -, (b) NO-, and (c) NO_2 -enriched NH_3/air flames by employing the decoupling approach detailed in Fig. 1(b).

Notably, at $x_a = 1.0$, NH_3 slip becomes nearly negligible over the range $\phi = 0.6$ –1.7. Comparing the effects of different nitrogen oxide species, it is observed that NO_2 addition leads to the highest NO emissions under lean conditions, as well as the largest NH_3 and H_2 slip under fuel-rich conditions. N_2O addition plays a comparatively minor role, though it still contributes to a moderate increase in NO emissions compared to NH_3/air flames. The effect of NO addition lies between the other two cases, showing intermediate levels of NO formation and fuel slip.

Following the decoupling methodology detailed in Section 2.4, the physical-chemical effects on NO formation associated with reactive oxidizer addition are examined in Fig. 13 at two representative equivalence ratios, $\phi = 0.7$ and $\phi = 1.4$. In all cases, the transport effects are found to be negligible, indicating that NO formation is primarily governed by thermal and chemical (including other) effects.

For cases of N_2O addition, the enhanced NO formation in the downstream region is shown to be predominantly driven by thermal effects. Under lean conditions, thermal contributions are approximately three times larger than the chemical & other effects. At rich conditions, e.g., $\phi = 1.4$, the chemical effects become almost negligible, further highlighting the dominant role of temperature in NO production in the presence of N_2O .

In contrast, for NO- or NO_2 -enriched flames, a significant portion of NO originates directly from the premixtures or from the decomposition of the added NO_2 . Consequently, the relative contribution of thermal effects is reduced, especially under lean conditions where the initial NO or NO_2 concentration is high, as illustrated in Fig. 13(b)–(c). It is also observed that at $\phi = 0.7$, thermal effects tend to promote NO formation at low levels of NO or NO_2 addition, but can become inhibitory as the x_a increases. Nevertheless, the chemical & other effects are the primary NO formation mechanisms. Under rich conditions, the dual role of thermal effects persists for NO addition, and the contribution of chemical effects is significantly reduced. For NO_2 addition at $\phi = 1.4$,

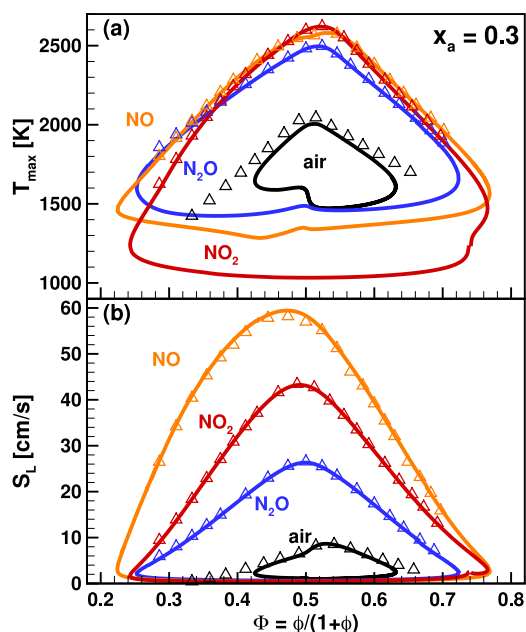


Fig. 14. Change of (a) maximal flame temperature T_{max} and (b) laminar flame speed S_L with respect to the normalized equivalence ratio $\Phi = \phi/(1+\phi)$ under radiative (lines) and adiabatic (symbols) conditions for $x_a = 0.3$.

thermal effects once again emerge as the dominant contributor to NO formation, while chemical effects become negligible.

3.3. Influence of radiation and flammability limits

In this section, radiative heat losses are taken into account using the optically thin radiation model [34]. Fig. 14 presents the variations in maximal flame temperature T_{max} and laminar flame speed S_L as functions of ϕ under both radiative and adiabatic conditions for different nitrogen oxide species, with x_a fixed at 0.3. Compared to adiabatic conditions, radiative heat losses lead to a slight reduction in maximal flame temperature near stoichiometric conditions. Their effects on the flame are weak when there are intense reactions, since they mainly lower the temperature in the post-flame zone. As ϕ deviates further from unity, the flame temperature and heat release rate drop, and radiation losses become more significant. Under radiative conditions, no flame solutions are found beyond certain equivalence ratios; these boundary values define the flammability limits of the mixture under the specified thermodynamic conditions, e.g., 298 K and 1 atm here. Note that no such limit exists when assuming adiabatic conditions. Overall, the incorporation of N_xO_y increases both flame temperature and extends the flammable region. For instance, no flame solution exists at $\phi = 0.5$ ($\Phi = 0.333$) or $\phi = 2.0$ ($\Phi = 0.667$) for NH_3/air mixtures, whereas stable flames are sustained when N_xO_y are present in the oxidizer stream.

Fig. 14(b) further compares the laminar flame speed under adiabatic and radiative conditions. It is observed that the impact of radiation on flame speed is relatively weak except near the lean and rich flammability limits, as evidenced by the close overlap between the adiabatic and radiative curves across $\phi = 0.5$ – 2.0 . Besides, we also assess the influence of radiation on flame speed at different x_a by comparing the flame speeds under radiative and adiabatic conditions at $\phi = 0.7$ and 1.4 . It is observed that the relative differences are minimal, suggesting that radiation has a negligible influence on flame speed across all conditions, as long as the equivalence ratio remains sufficiently far from the flammability limits. These have been included in the supplementary materials for reference.

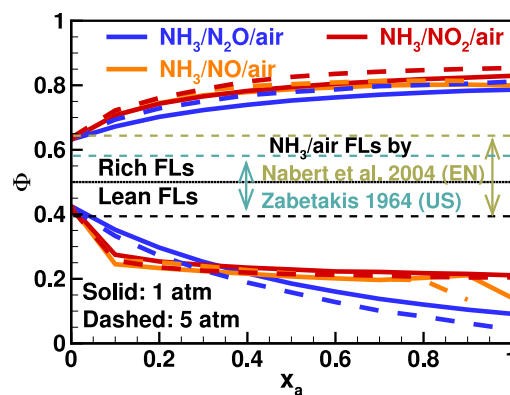


Fig. 15. Change of flammability limits with x_a for $\text{NH}_3/\text{N}_2\text{O}/\text{air}$, $\text{NH}_3/\text{NO}/\text{air}$, and $\text{NH}_3/\text{NO}_2/\text{air}$ mixtures at 1 and 5 atm.

Fig. 15 compares the flammability limits derived from unstretched planar premixed flames for different N_xO_y additions, where the normalized equivalence ratio $\Phi = \phi/(1+\phi)$ rather than fuel molar fractions is considered. The calculated flammability limits for the baseline NH_3/air mixture are 0.740 on the lean side and 1.715 on the rich side. For reference, experimental data from Refs. [39–41] are shown as dashed lines. It is observed that the computed lean flammability limit is slightly higher than those reported experimentally—0.63 by Zabetakis [39] and 0.65 by Nabert et al. [40]. On the rich side, the present result (1.715) is closer to the value of 1.807 reported by Nabert et al. while exceeding the value of 1.389 from Zabetakis. Overall, the numerical predictions demonstrate reasonable agreement with experimental observations.

The influence of reactive oxidant addition on flammability limits is further analyzed. Overall, the inclusion of N_xO_y into the oxidizer stream significantly broadens the flammability range. The N_2O addition typically leads to a gradual and continuous extension of both the lean and rich flammability limits as its concentration increases. In contrast, for NO and NO_2 addition, on the lean side, a relatively low addition of NO or NO_2 (e.g., $x_a = 0.1$) can lead to a substantial expansion of the flammability limit—from $\phi = 0.740$ to 0.325 for NO and 0.379 for NO_2 , respectively. However, further increases in x_a , e.g., $x_a \geq 0.2$, result in only marginal additional benefits, except in the NH_3/NO mixtures. These indicate that at lower x_a , NO and NO_2 are more effective in extending the lean flammability limit, compared to N_2O . Nonetheless, when $x_a \geq 0.5$, N_2O is superior in extending the flammable range. An interesting observation is that NO addition causes a sudden jump in the lean flammability limit at $x_a = 1.0$, which is related to its high-temperature thermal decomposition. For $x_a \leq 0.8$, decreasing ϕ lowers the flame temperature and slows the decomposition of NO . A significant quantity of NO remains undecomposed, and the lean flammability limit is predominantly determined by the oxidation of NH_3 . However, at $x_a = 1.0$, the elevated flame temperature is beneficial for the decomposition of NO , and the heat emitted from NO decomposition (into N_2 and O_2) further supports this process. As a result, extinction is governed by the synergistic effects of NH_3 consumption and NO decomposition. This leads to a sharp change in the lean flammability limit for $\text{NH}_3/\text{NO}/\text{air}$ mixtures. More supporting information can be found in the Supplementary Materials for interested readers.

On the rich side, a trend similar to that observed for $\text{NH}_3/\text{N}_2\text{O}/\text{air}$ mixtures is found for all three N_xO_y species, characterized by a steady rise in the upper flammability limit with x_a . This is attributed to the increasing availability of oxygen from these additives, which becomes the limiting factor under fuel-rich conditions. In general, among the three oxidizers, NO_2 exhibits the strongest strengthening effect, followed by NO and N_2O .

Fig. 15 also contains results at an elevated pressure of 5 atm. Increasing the pressure only slightly broadens the flammability limits on both the lean and rich sides. The general trends regarding

x_a for different N_xO_y addition remain largely unchanged, indicating that the conclusions drawn from atmospheric conditions continue to hold at higher pressures. Furthermore, a notable observation under high pressure is that NO-enriched flames exhibit a sharp reduction in the lean flammability limit at high x_a , around 0.9. This behavior is attributed to the enhanced NO decomposition under high-pressure and high-temperature conditions.

4. Conclusions

This study presented a comparative numerical investigation of laminar premixed ammonia/air flames enriched with nitrogen oxide species, specifically N_2O , NO, and NO_2 , to advance the fundamental understanding of ammonia oxidation chemistry. The effects of N_2O , NO, and NO_2 addition on laminar flame speed, pollutant emissions, and flammability limits were systematically analyzed.

The results reveal that all three nitrogen oxide species significantly enhance flame propagation. The oxy-enrichment effects were first isolated, showing a progressive increase in influence from N_2O to NO and then to NO_2 . Then, the oxy-enrichment effect is further decoupled into the thermal, transport, and chemical effects. The analyses show that thermal and chemical effects are the two primary effects related to N_xO_y addition, while the transport effects are negligible. Specifically, N_2O primarily enhances flame speed thermally, with a weak and slightly negative chemical influence. In contrast, NO and NO_2 exhibit more pronounced chemical contributions: NO_2 consistently enhances flame propagation chemically, while NO has a dual effect—promoting combustion at low concentrations but inhibiting it at higher levels. Then, a detailed chemical kinetic analysis was carried out, incorporating rate-of-progress analysis, reaction network analysis, and rate constant sensitivity analysis, to identify the key reactions. The results show that the strong chemical contributions in NO- or NO_2 -enriched flames are primarily related to interactions between NH_i radicals and NO species. Meanwhile, the correlations between the simulated laminar flame speeds and the theoretical expressions are examined over different addition ratios. While the theoretical expression serves as a useful baseline, its predictive accuracy diminishes in regimes with complex chemical kinetics.

The pollutant emissions, especially NO and unburnt NH_3 , are examined downstream of the flame. In all cases, the addition of nitrogen oxides tends to increase pollutant levels. For N_2O addition, the thermal effect is the dominant factor influencing emissions, particularly at rich conditions. However, for NO or NO_2 additions, chemical effects play a more significant role. Additionally, a dual thermal effect is observed under lean conditions.

Moreover, the radiation is taken into account via the optically thin model and is found to have a negligible influence on flame speed except near the flammability limits. The flammability limits are numerically estimated, and it is demonstrated that N_2O and NO_x addition can substantially broaden the flammable range of ammonia/air mixtures. In particular, NO and NO_2 are effective in extending the lean flammability limit even at low concentrations. However, when x_a ratio exceeds 0.2, it could only result in a marginal extension of lean flammability limits. In contrast, N_2O addition leads to a more gradual and sustained widening of the flammability range with the increase of x_a .

In this study, we adopted a single-species enrichment approach to systematically isolate and compare the influence of each individual N_xO_y species on NH_3 /air flames. In practical combustion systems, however, the coexistence of multiple nitrogen-containing oxidizers at different relative concentrations is more common. Future studies that consider simultaneous enrichment by different N_xO_y species would provide more comprehensive insights. Note that experimental data for $NH_3/N_2O/NO_x$ flames remain rather limited, and future experimental work on this subject would be highly beneficial for model development and validation.

CRediT authorship contribution statement

Shumeng Xie: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peng Ma:** Writing – review & editing, Investigation, Formal analysis. **Jinzhou Li:** Writing – review & editing, Investigation, Formal analysis. **Samir Boset Rojas Chávez:** Writing – review & editing, Investigation, Formal analysis. **Surendra Kumar Soni:** Writing – review & editing, Investigation, Formal analysis. **Hao Hu:** Writing – review & editing, Investigation, Formal analysis. **Huangwei Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.combustflame.2025.114518>.

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